REASONING AND PLANNING

Reasoning is the ability of an intellect, artificial or not, to understand the process of thought or thinking through a problem. An intellect can understand that actions have outcomes, and it can use this ability to reason through which action from a set of actions can be applied to solve a given task.

Planning is the ability of the intellect to reason out the order of actions or tasks and apply the correct parameters to achieve a goal or outcome—the extent to which an intellectual plan depends on the scope of the problem. An intellect may combine multiple levels of planning, from strategic and tactical to operational and contingent.

We'll look at another set of prompt engineering techniques that allow or mimic reasoning behavior to demonstrate this reasoning ability. Typically, when evaluating the application of reasoning, we look to having the LLM solve challenging problems it wasn't designed to solve. A good source of such is based on logic, math, and word problems.

Using the time travel theme, what class of unique problems could be better to solve than understanding time travel? Figure 10.3 depicts one example of a uniquely challenging time travel problem. Our goal is to acquire the ability to prompt the LLM in a manner that allows it to solve the problem correctly.